4.5.6 Poly-phase AC generator

Some AC generators may have more than one coil in the armature core and each coil produces an alternating emf. In these generators, more than one emf is produced. Thus they are called **poly-phase generators**.

If there are two alternating emfs produced in a generator, it is called two-phase generator. In some AC generators, there are three separate coils, which would give three separate emfs. Hence they are called **three-phase AC generators**.

4.5.7 Three-phase AC generator

In the simplified construction of three-phase AC generator, the armature core has 6 slots, cut on its inner rim. Each slot is 60° away from one another. Six armature conductors are mounted in these slots. The conductors 1 and 4 are joined in series to form coil 1. The conductors 3 and 6 form coil 2 while the conductors 5 and 2 form coil 3. So, these coils are retangular in shape and are 120° apart from one another (Figure 4.30).

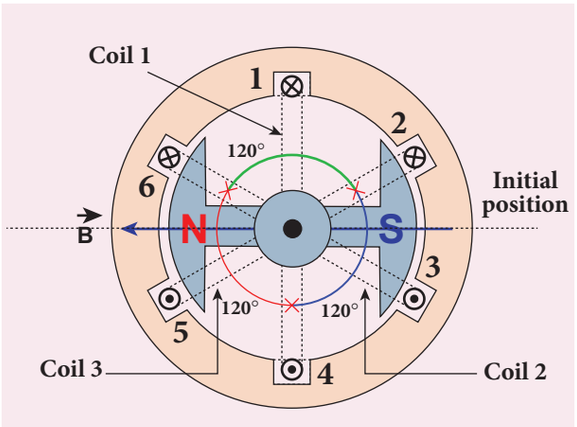


Figure 4.30 Construction of three-phase AC generator

The initial position of the field magnet is horizontal and field direction is perpendicular to the plane of the coil 1. As it is seen in single phase AC generator, when field magnet is rotated from that position in clockwise direction, alternating emf ε_1 in coil 1 begins a cycle from origin O. This is shown in Figure 4.31.

The corresponding cycle for alternating emf ε_2 in coil 2 starts at point A after field magnet has rotated through 120°. Therefore, the phase difference between ε_1 and ε_2 is 120°. Similarly, emf ε_3 in coil 3 would begin its cycle at point B after 240° rotation of field magnet from initial position. Thus these emfs produced in the three phase AC generator have 120° phase difference between one another.

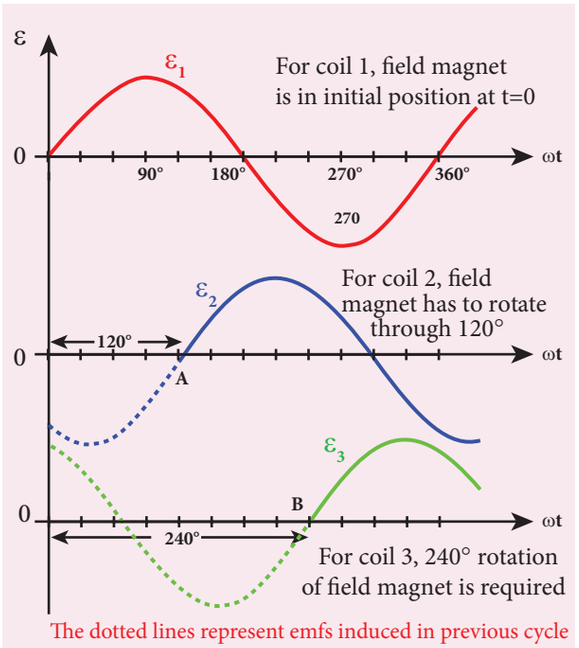


Figure 4.31 Variation of emfs ε_1, ε_2 and ε_3 with time angle.

4.5.8 Advantages of three-phase alternator

Three-phase system has many advantages over single-phase system. Let us see a few of them.

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